Homework 7

Due date: Jun.3^{rd}

Turn in your homework in class

Rules:

- Please work on your own and submit **on time**. Discussion is permissible, but extremely similar submissions will be judged as plagiarism!
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain **3** decimal places if rounding is required without special requirements for the question,
- Please staple your homework before submitting; take a photo and upload it on time for online record, including the grade (with any possible deductions) after receiving the marked copy.

1

[12 points]Draw the Bode plot of the following transfer function.
(a)$H(\omega) = \frac{j\omega+10}{jw(j\omega+5)^2}$
(b)$H(\omega) = \frac{j\omega+1}{-\omega^2+20j\omega+100}$

2

[14 points]The circuit is shown in **Figure 2**, find:
(a)The transfer function $H(\omega) = \frac{V_o(\omega)}{I(\omega)}$;
(b)The magnitude of H at $\omega_0 = 1$ rad/s.

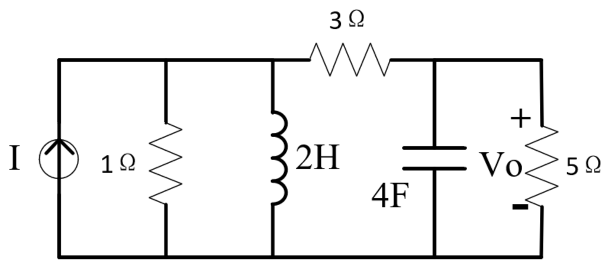

Figure 2

3

[10 points] For the circuit in **Figure 3**, find the resonant frequency ω_0 and $Z_{in}(\omega_0)$.

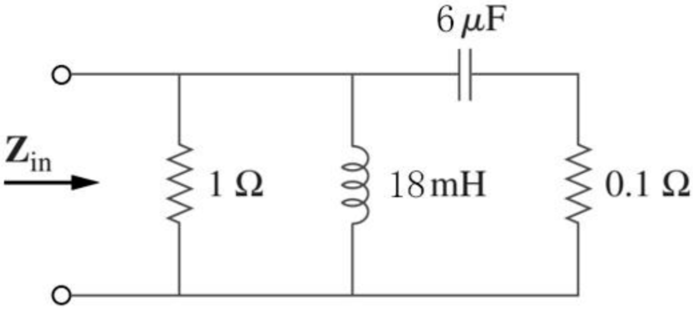

Figure 3

4

[21 points] Consider the circuit in **Figure 4**. The circuit is a parallel RLC circuit

(a) Determine the resonant frequencies, ω_0 (rad/s).

(b) Find the Q of the circuit.

(c) Calculate the voltage **V** across the circuit at resonance.

(d) Determine the bandwidth of the circuit in rad/s.

(e) Sketch the magnitude of **V** as a function of frequency ω, indicating the voltage at the half-power frequencies.

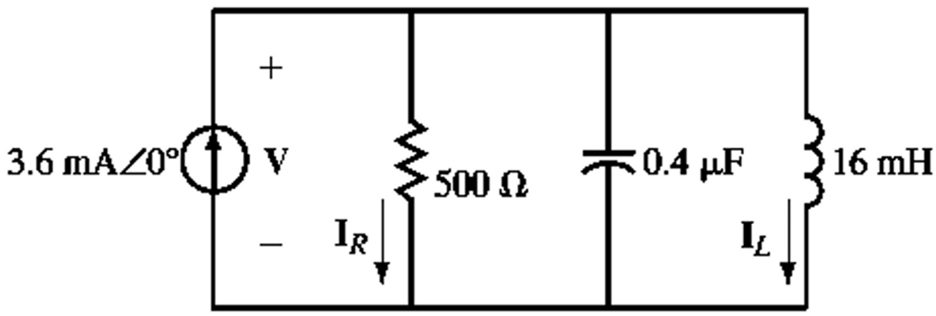

Figure 4

5

[15 points] Determine the type of the following filters. Find the bandwidth and the center frequency of the following filters.

(a) The transfer function of an example filter is

$$H(\omega) = \frac{j\omega K_1}{(j\omega)^2 + j\omega K_1 + K_2^2}$$

where $K_1 > 0$ and $K_2 > 0$.

(b) The transfer function of an example filter is

$$H(\omega) = \frac{(j\omega)^2 + K_2^2}{(j\omega)^2 + j\omega K_1 + K_2^2}$$

where $K_1 > 0$ and $K_2 > 0$.

(c) The transfer function is $H(\omega) = \frac{\mathbf{V}_o(\omega)}{\mathbf{V}_s(\omega)}$ for the circuit in **Figure 5**. (You may refer to the above two questions.)

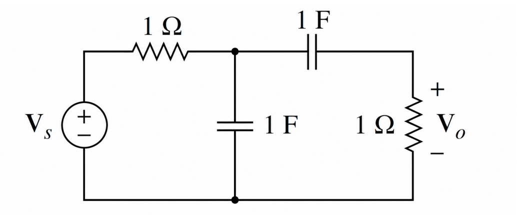

Figure 5

6

[12 points] A second-order active filter is shown in **Figure 6**. The operational amplifier is ideal and operates in the linear region.

(a) Express the transfer function $H(\omega) = \mathbf{V_o}/\mathbf{V_i}$ using C_1, C_2, R_1 and R_2.

(b) What kind of filter is it? Why?

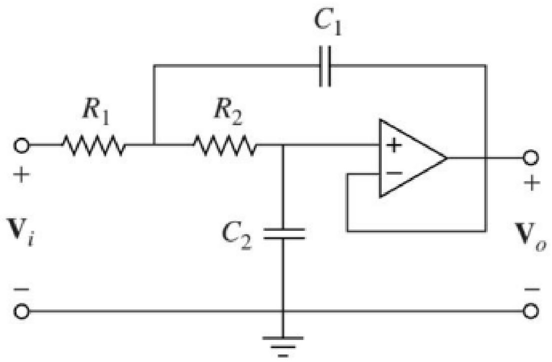

Figure 6

7

[16 points] Suppose that $R = 1\,\mathrm{k\Omega}$ and $R_i = 2\,\mathrm{k\Omega}$. Define the input signal as $\mathbf{V_i}(\omega)$ and the output signal as $\mathbf{V_o}(\omega)$. The transfer function $H(\omega) = \frac{\mathbf{V_o}(\omega)}{\mathbf{V_i}(\omega)}$. The operational amplifiers are ideal and operate in the linear region. Design an active band-reject filter according to **Figure 7** with the following specifications.

(a). Gain: $K = 30$

(b). The cutoff frequency of the low-pass filter: $f_1 = 10\,\mathrm{kHz}$

(c). The cutoff frequency of the high-pass filter: $f_2 = 50\,\mathrm{kHz}$

Determine C_1, C_2, R_f, quality factor Q, and the expression for the transfer function $H(\omega)$..

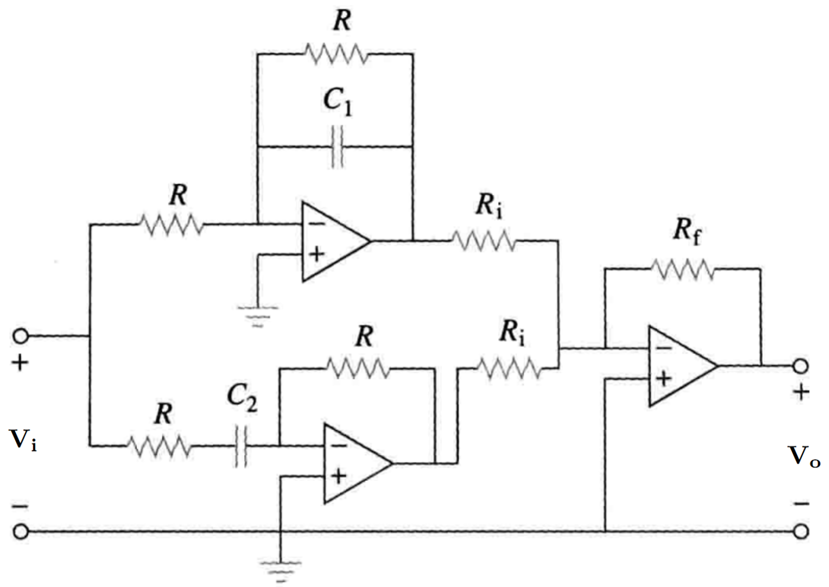

Figure 7